ENGINEERING

AI-assisted optoelectrokinetic control of active self-propelling micromotors for independent navigation with multimode motions

Jiaxin Liu[1]†, Zhiqiang Zheng[2,3]†, Yaozhen Hou[1], Qing Shi[4], Qiang Huang[5], Jie Han[3], Metin Sitti[3,6]*, Huaping Wang[4]*

Agile and controllable motion is essential for active micromotors to navigate various microscale scenarios and manipulate the microscopic world, where parallel navigation and independent control are highly desirable. However, most micromotor navigation strategies suffer from limited agility and poor predictability, with motion typically confined to single-micromotor self-propulsion along predetermined trajectories. Herein, we propose an artificial intelligence (AI)–assisted optoelectronic control strategy that involves converting stochastic micromotor self-propulsion into controllable omnidirectional motion by synergistically exploiting multiple electrokinetic mechanisms. This strategy enables independent navigation of individual micromotors while simultaneously supporting parallel manipulation. By spatiotemporally configuring two or more optical patterns, agile motion primitives, such as directional propulsion, passive propulsion, in situ U-turns, and motion pause and restarting, were developed. To improve navigation robustness under coupled electrokinetic effects, a spatial-temporal AI model was developed for accurately predicting micromotor motion to facilitate the optimization of dynamic guidance schemes. These motion primitives are sequentially integrated and automatedly switched along long-term, reconfigurable trajectories, thereby enabling continuous navigation guided by discrete optical patterns. Independent control of active micromotors was demonstrated through the parallel manipulation of multiple Janus micromotors that were navigating intricate networks, in which each micromotor followed individual trajectories and adapted in real time to local terrain variations. This work showcased an agile and predictable navigation strategy for active micromotors that facilitates independent and massively parallel manipulation in intricate terrains, thus opening further possibilities for advanced applications.

INTRODUCTION

Living microswimmers, such as paramecium, sperm, motile algae, and bacteria, exhibit autonomous motion (*1*, *2*) and self-navigation behaviors that are actuated by biological energy (*3*) and environmental signals (*4*). Inspired by these natural systems, artificial active micromotors have been constructed (*5*). Typically featuring anisotropic structures or heterogeneous compositions (*6*, *7*), micromotors enable self-propulsion via asymmetric conversion of ambient energy (*8*, *9*), even without external physical or chemical gradients. Owing to their tiny size, functional versatility, cargo compatibility (*10*–*12*), and ability to harness environmental energy, active micromotors hold considerable promise for applications (*13*–*15*) such as cargo transport (*16*, *17*), environmental remediation (*18*, *19*), microfabrication (*20*), and even thermodynamic studies. The ideal motion of active micromotors are expected to have high maneuverability and precision controllability, thereby enabling rapid navigation to designated regions, agile steering, and reliable stopping as will. However, thermal fluctuations (*21*) induce stochastic and time-varying propulsion directions of active micromotors, thus hindering the robust and efficient navigation of multiple micromotors. To modulate self-propulsion behavior and navigate one or multiple micromotors within intricate environments, control strategy with high predictability and agility is urgently needed (*22*).

Several strategies have been proposed for steering active micromotors. Signals such as chemical gradients (*23*), light gradients (*24*, *25*), light directions (*26*), and even fluid flow (*27*) can be used to guide micromotors toward or away from the signal source, which is generally referred to as taxis-based strategy (*28*). However, taxis behavior occurs only when micromotors exhibit special properties, such as chemotaxis (*29*) and phototaxis (*30*), and is further limited by the low sensitivity of micromotors to such signal cues (*31*). Alternatively, the geometric boundaries of solid photoresists (*32*–*34*) and patterned microelectrodes (*35*) can guide the motion directions of micromotors without physical or chemical gradients. However, these microstructures should be predesigned and premanufactured (*36*) in advance, which is a tedious and inherently nonreconfigurable process. Magnetic fields (*37*) enable flexible orientation control of magnetic micromotors but are deficient in independent control of individual micromotors because of the identical responses of all microparticles under the same magnetic excitation (*38*). Acoustic fields (*39*, *40*) can generate localized circular microstreaming, thus endowing micromotors with tunable torques and independent rotation. Nevertheless, achieving agility, precision, and independent navigation of multiple micromotors remains notably challenging.

More recently, optoelectronic techniques (*41*) have emerged as a promising route for parallel manipulation of microbeads and for guiding self-propulsion of individual micromotors via induced charge

[1]Intelligent Robotics Institute, School of Mechatronical Engineering, Beijing Institute of Technology, Beijing 100081, China. [2]Department of Biomedical Engineering, City University of Hong Kong, 999077 Hong Kong, China. [3]Physical Intelligence Department, Max Planck Institute for Intelligent Systems, Stuttgart 70569, Germany. [4]Key Laboratory of Biomimetic Robots and Systems (Beijing Institute of Technology), Ministry of Education, Beijing 100081, China. [5]Department of Robotics and Autonomous Systems, The Hong Kong University of Science and Technology (Guangzhou), Guangzhou 511453, China. [6]School of Medicine and College of Engineering, Koç University, 34450 Istanbul, Turkey.
*Corresponding author. Email: sitti@is.mpg.de (M.S.); wanghuaping@bit.edu.cn (H.W.)
†These authors contributed equally to this work.

electrophoresis (ICEP) (*42*, *43*). However, existing optoelectronic approaches typically confine micromotors within optical patterns, which substantially limits motion versatility and flexibility, with the notable absence of in-plane rotation and propulsion away from optical patterns (*43*). Furthermore, the motion of active micromotors within an optoelectronic device is governed by coupled effects of multiple electrokinetic mechanisms, including ICEP, dielectrophoresis (DEP), and alternating current electroosmosis (ACEO). The resulting forces are highly spatial dependence and are challenging to quantify and decouple analytically. In addition, local electric fields, which are nonlinearly shaped by multiple optical patterns or even distinct regions within a single optical pattern, complicate the precision guidance of self-propulsion. Thus, accurate motion prediction under coupled optoelectronic effects is essential for achieving agile, independent, and controllable navigation of multiple micromotors.

To address this challenge, we present an optoelectronic control strategy that incorporates artificial intelligence (AI)–based prediction to investigate the motion behaviors of active micromotors and to develop stable and reliable guidance schemes for micromotor navigation. This strategy improves motion agility by comprehensively leveraging the spatial inhomogeneity inherently that is associated with electrokinetic effects including ICEP, DEP, and ACEO (Fig. 1A). As a result, omnidirectional motion encompasses propulsion along the tangential direction of the optical pattern, propulsion along the vertical direction, and in-plane rotation is achieved. Moreover, by spatially and temporally configuring two or more optical patterns, multiple motion primitives of active micromotors, including directional propulsion, passive propulsion, in situ U-turns, and motion pause and restart were developed. To improve long-term propulsion, these motion primitives were selectively and sequentially integrated along distinct trajectories and were modularly reconfigurable as required. The dynamic behavior, both transient and long-term of micromotors, could be accurately predicted by an AI-based model, thus providing a cost-free and high-efficiency platform for testing and optimizing of optical pattern parameters. For parallel and independent navigation of multiple micromotors, optical patterns were automatedly reconfigured as micromotors propelled forward, with the assistance of visual perception. Multiple micromotors self-propelled within intricate optoelectronic networks, thus showcasing independent control and in-time adaptation to environmental changes. Our proposed AI-assisted optoelectronic control strategy is potentially applicable to accurate prediction and automated control of other microswimmers under coupled actuation effects.

RESULTS

Self-propelling motion induced by a 1D optoelectronic pattern

A metallic-colloidal Janus micromotor consists of a polystyrene (PS) colloidal hemisphere and a metallic hemisphere. Such micromotors were fabricated by sequentially evaporating 10-nm Cr and 10-nm Au onto one side of PS microspheres. Under an AC electric field, the polarization contrast between the metallic and colloidal hemispheres of the Janus micromotor generated an asymmetric electroosmotic flow, which enabled self-propulsion of the micromotor toward its colloidal hemisphere via ICEP. According to the ICEP mechanism, the propulsion direction coincides with the normal direction of the Janus interface, which points toward the colloidal hemisphere. By projecting optical patterns onto a hydrogenated amorphous silicon (a-Si:H) photoconductive layer coated on an indium tin oxide (ITO) glass slide, the illuminated regions exhibited markedly higher conductivity than the dark regions did, thus generating a spatially nonuniform electric field. This electric field guided the self-propelling micromotors along directions that minimized the internal energy of micromotors. In addition to ICEP, DEP and ACEO also contributed to the micromotor motion in the optoelectronic system (Fig. 1A). When the dielectric properties of the colloidal hemisphere dominated, a negative dielectrophoretic (n-DEP) force repelled the micromotors from illuminated regions. Moreover, ACEO flow, which occurred mainly near the edges of the optical patterns, tended to propel the micromotors from dark areas toward illuminated areas. ACEO flow is often regarded as a source of interference in directional steering. In this study, however, we found that ACEO flow induced micromotor motion with a nontangential component relative to the edges of optical pattern. The finite element analysis (FEA) results for both the negative n-DEP force and the ACEO flow field are shown in Fig. 1 (B and C). The n-DEP force was dominant when the micromotor was far from the optical pattern, whereas the ACEO became dominant at short distances (Fig. 1D). At a specific location offset from the optical patterns, these two forces were counterbalanced, thus yielding a steady-state velocity parallel to the optical-pattern edge. Figure 1E further elucidates the attraction mechanism caused by ACEO, the repulsion mechanism actuated by n-DEP, and the planetary-like motion behavior that arose from the interplay of these effects are further elucidated in Fig. 1E.

The motion behaviors of metallic-colloidal micromotors induced by one-dimensional (1D) (spot-like and linear) optoelectronic patterns in AC electric fields were comprehensively investigated. As shown in Fig. 2A, the micromotors exhibited planetary-like self-propulsion motion around the spot-like optical patterns and traced circular trajectories with specific rotational radii. DEP and ACEO maintained the rotation radius, whereas the ICEP effect primarily regulated the tangential component of motion. The orbital radius increased linearly with the increasing optical pattern diameter over the range of 1.5 to 12 μm, whereas the angular velocity increased over the range of 1.5 to 7.5 μm. Beyond a diameter of 12 μm, these two parameters were saturated at approximately 28 μm (Fig. 2B) and 70°/s (Fig. 2C), respectively. Varying the optical-pattern diameter shifted the relative strengths of the DEP, ACEO, and ICEP effects, thereby modulating the rotational radius and angular velocity. The motion of Janus micromotors near linear optical patterns can be divided into three states: attraction, fluctuation, and linear propulsion (Fig. 2D). Initially, micromotors were attracted from a distance toward the light pattern; subsequently, they exhibited brief positional fluctuations along the axis perpendicular to optical patterns; last, steady linear self-propulsion was achieved. The ACEO effect dominated the attraction state; thus, the displacement of each micromotor had both horizontal and vertical components. As demonstrated in Fig. 2E, the vertical component of the displacement increased proportionally to the distance between the micromotor's initial position and the optical pattern. Furthermore, the velocity of the micromotor increased markedly during attraction, owing to the strengthening ACEO and ICEP forces as it approached the light patterns. The critical attraction distance increased with increasing optical-pattern width and stabilized at ~176 μm (Fig. 2F). Micromotors initially positioned beyond this distance exhibited no measurable attractive response. In the fluctuation state, the vertical displacement of the micromotors oscillated over time because of out-of-plane rotation of the Janus interface. We

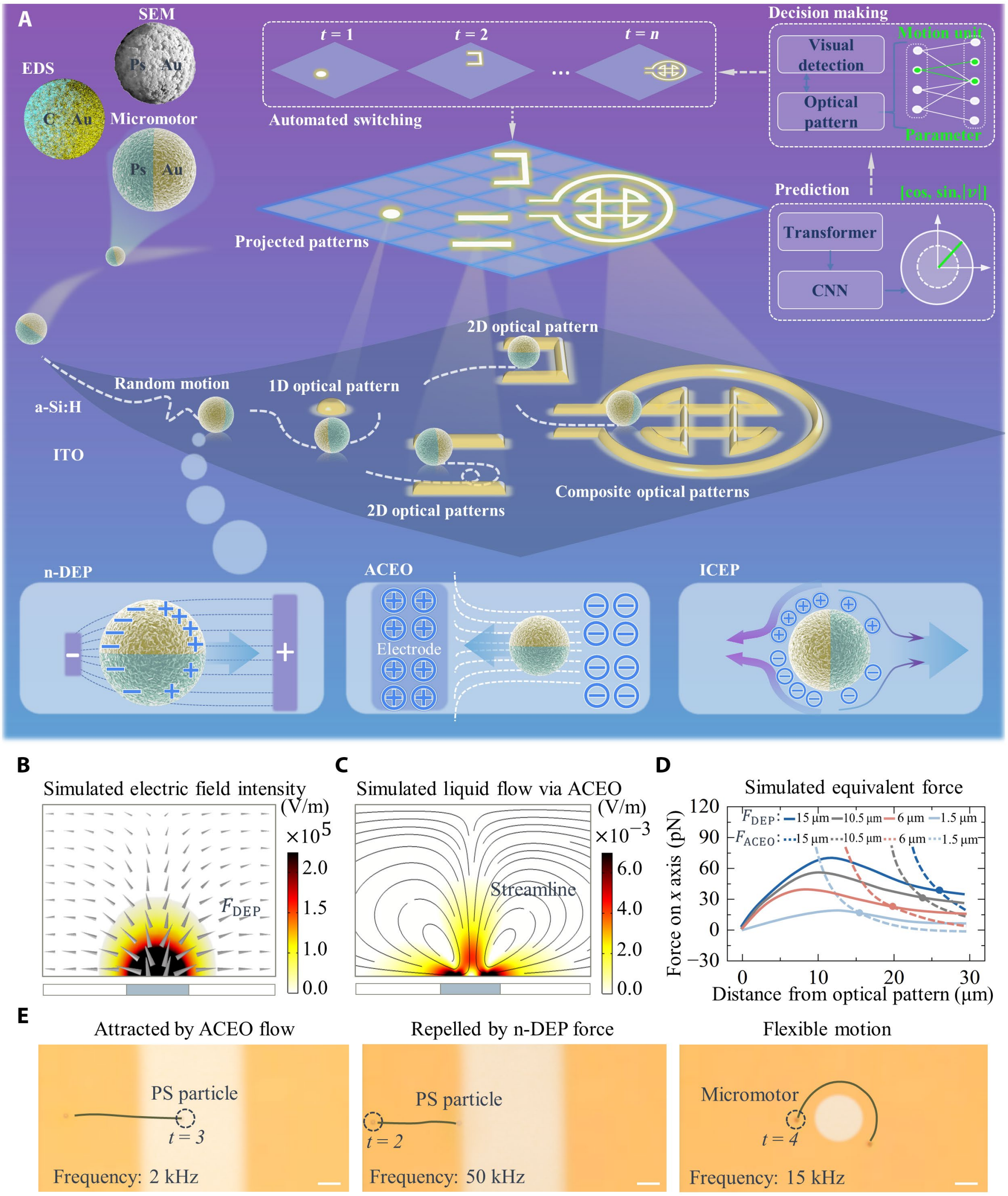

Fig. 1. Micromotor motion affected by combination effects of three electrokinetic mechanisms. (**A**) Active micromotor navigation guided by reconfigurable optical patterns. Janus micromotors with Au and PS hemispheres move randomly without external constraints but become agile and controllable under guidance and constraints from optical patterns. AI-based motion prediction and decision-making enhance the robustness of navigation. FEA of electrokinetic mechanisms: (**B**) n-DEP repels micromotor from optical patterns, and (**C**) ACEO attracts micromotor toward optical pattern. (**D**) Numerical simulation of n-DEP force and ACEO force versus the distance from optical patterns. (**E**) Experimental images of microparticle motion predominantly induced by ACEO, n-DEP, and ICEP effects. Scale bar, 20 μm.

quantified this state using two metrics: overshoot and decay rate. Overshoot was defined as the peak deviation from the maximum vertical position relative to the offset distance (d_5/d_6), whereas decay rate was defined as the amplitude ratio of the second to the first fluctuation cycle [$(d_3 + d_4)/(d_1 + d_2)$]. The average values of both metrics were positively linearly correlated with the light pattern width below the thresholds of 9 μm (Fig. 2G) and 13.5 μm (Fig. 2H), respectively. In the linear moving state, the Janus interface of the micromotor was oriented perpendicular to the edges of the light pattern after self-alignment. This alignment resulted in a relatively stable propulsion velocity and a consistent displacement offset. As shown in Fig. 2 (I and J), the offset distance (d_6) and velocity were both positively and linearly correlated with the light pattern width under these conditions. However, the influence of width on the offset distance d_6 decreased as the width increased further. Representative trajectories of four micromotors guided by optical patterns of diverse

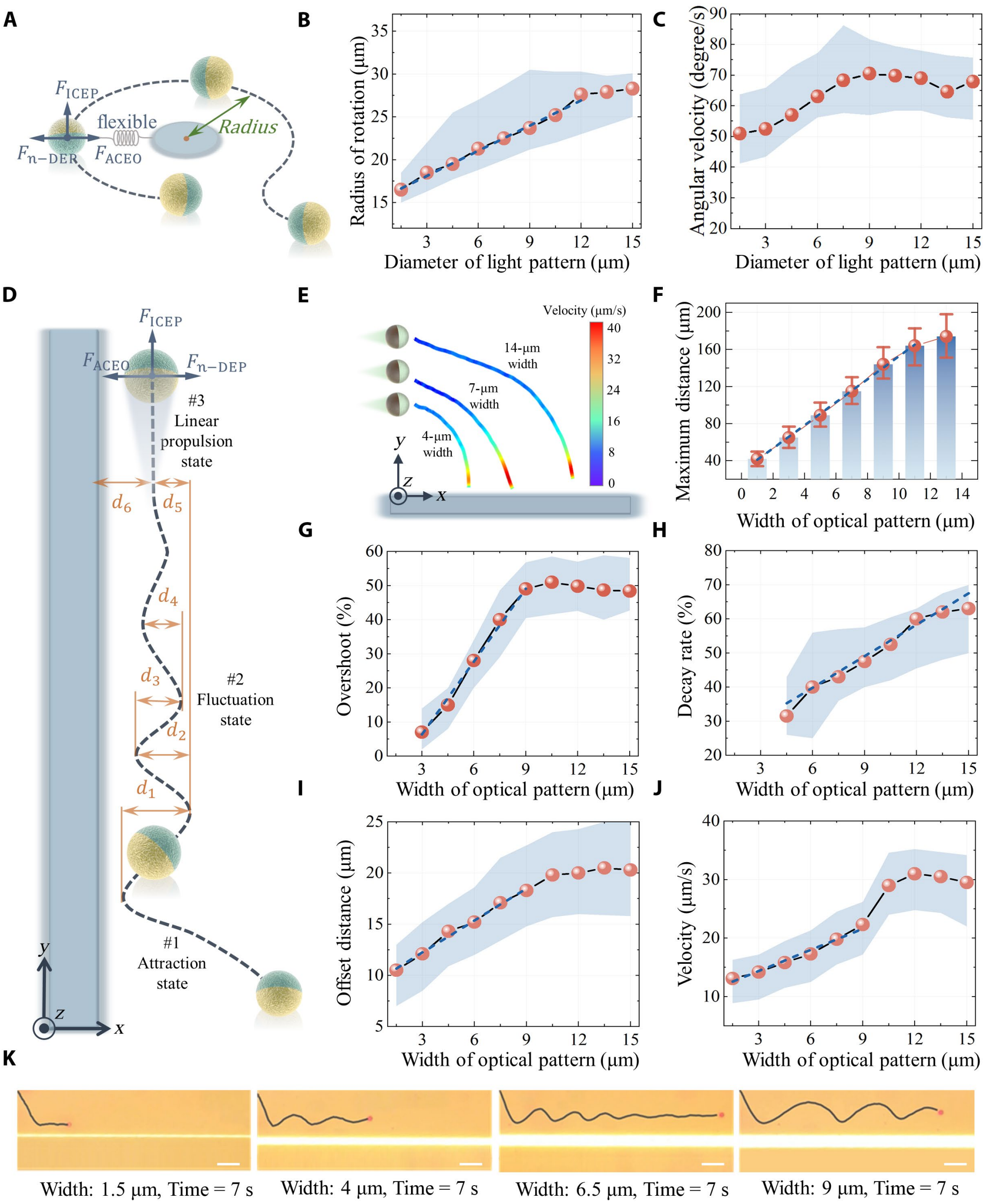

Fig. 2. Micromotor motion induced by single optical pattern. (**A**) Planetary motion of micromotors around spot-like optical pattern. (**B**) Rotation radius and (**C**) angular velocity of planetary motion versus diameter of spot-like light patterns (**D**) Motion behaviors of micromotors around a linear light pattern, from attraction state, briefly fluctuating state, to linear propulsion state. (**E**) Trajectory and velocity of three micromotors being attracted by optical patterns with three diverse widths. (**F**) Critical distance from optical patterns for micromotors being attracted versus width of optical patterns. (**G**) Corelationship between overshoot and decay rate with width of light patterns in fluctuation state. (**H**) Corelationship between the decay rate with width of light patterns in fluctuation state. (**I**) Offset distances and (**J**) velocity in linear propelling state versus width of light patterns. (**K**) Trajectories of four micromotors guided by optical patterns with different widths. Scale bar, 20 μm.

width over 7 s are shown in Fig. 2K. In summary, these results indicate that: (i) ICEP does not always dominate micromotor motion; (ii) the coexistence of ICEP, ACEO, and DEP enables the regulation of both horizontal and vertical displacement as desired; and (iii) relative spatial positions significantly influence the motion behavior of micromotors.

Velocity prediction based on CNN-attention model

The propulsion velocity of each micromotors was determined primarily by the geometric configuration of the optical pattern, including the width, width gradient, and local curvature (Fig. 3A and movie S1). These parameters reshaped the local AC electric field and thereby modulated the coupled effects of ICEP, DEP, and ACEO. The

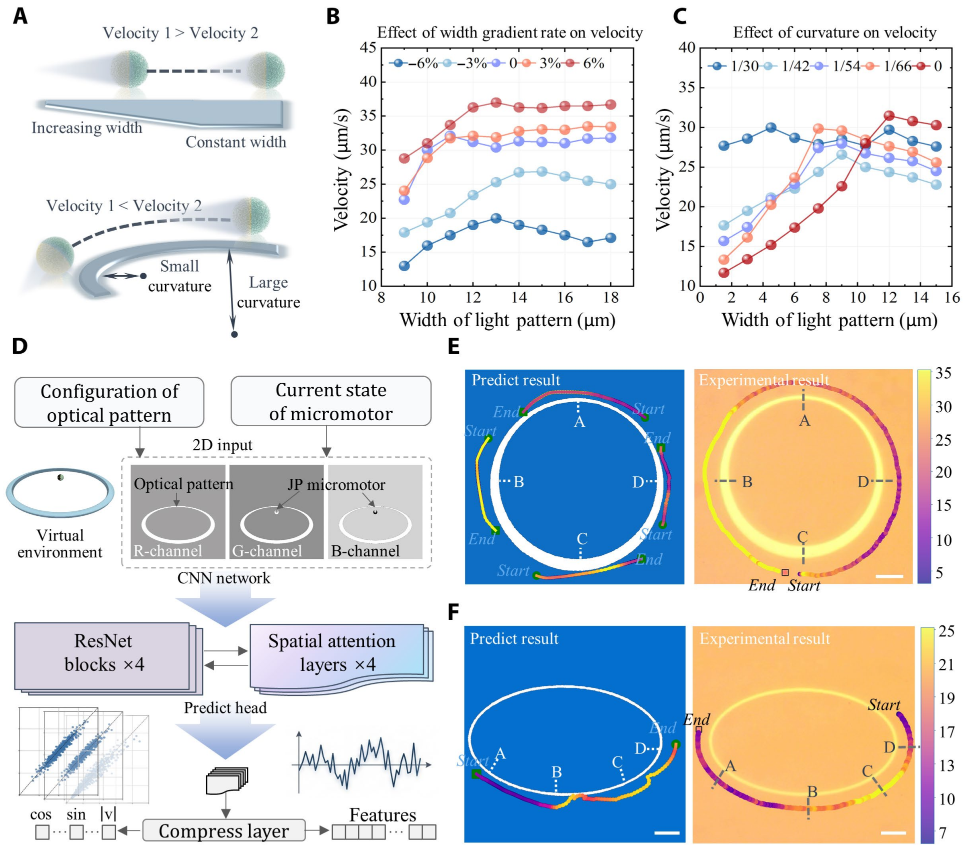

Fig. 3. Optical pattern geometry parameters modulate propulsion velocity. (**A**) The effects of width and curvature of optical patterns on self-propulsion velocity. (**B**) Optical-pattern-width–dependent velocity as a function of delta-width-length ratio varying from −6 to 6%. (**C**) Optical-pattern-width–dependent velocity as a function of curvature varying from 1/31 to 1/62 μm^{-1}. (**D**) Transient velocity prediction using a deep-learning model integrating the ResNet blocks and spatial attention. (**E**) Predicted and experimental velocity distribution of a micromotor along optical pattern with spatial-varying width. (**F**) Predicted and experimental velocity distribution of a micromotor along optical pattern with spatial-varying curvature. The predicted results are consistent with the experimental results. Scale bars, 20 μm.

width gradient (Δ*width*/Δ*length*, which is expressed as % per unit length) was strongly correlated with the propulsion velocity: Positive gradients accelerated motion, whereas negative gradients decelerated it. This relationship was quantified using experimental data on average velocity, gradient magnitude, and optical pattern width (Fig. 3B). The results demonstrated that the width gradient governed propulsion velocity over a broad parameter space. Conversely, the effect of width on propulsion velocity decreased as the magnitude of the width gradient increased. The velocity-curvature relationship was nonmonotonic and critically dependent on the width of the optical pattern (Fig. 3C). Specifically, for optical patterns with small widths (width <4.5 μm), velocity increased with curvature. However, for optical patterns with large width (width >10.5 μm), velocity decreased with the increasing curvature. The experimental results indicated that a high path curvature (small curvature radius) could also attenuate the modulatory effect of width on the propulsion velocity. A clear demonstration was observed at a curvature of 1/30 μm^{-1}, where the velocity remained almost invariant (27 to 32 μm/s) across a large range of pattern widths. Illumination intensity also affected the propulsion velocity, as depicted in fig. S1.

Several geometric parameters are just an incomplete characterization of optical-pattern features as they lack critical spatial information such as the micromotor's position relative to the pattern and its instantaneous state. To more comprehensively learn the relationship between the 2D optical-pattern features and micromotor dynamics, we developed a convolutional neural network (CNN) module with ResNet blocks (*44*) and a spatial attention (*45*) mechanism (Fig. 3D and table S1). The ResNet block enables the extraction of abstract features from 2D images and learning spatial configuration information from optical-pattern terrains, and the spatial attention mechanism emphasizes regions with greater influence on micromotor motion. The CNN model directly outputs both the direction and magnitude of the propulsion velocity through dedicated prediction heads. To evaluate the prediction efficiency, we compared the model-predicted velocities of a micromotor moving along two ring-shaped optical patterns with actual experimental velocities. The experimental and simulation results show strong concordance as follows: (i) For optical patterns with spatially varying widths, the micromotor reached its maximum velocity in region B (with increasing width) and its minimum velocity in region D (with decreasing width), as illustrated in Fig. 3E and movie S1. (ii) For optical patterns with spatial-varying curvatures, the micromotor moved fastest in region C (with decreasing curvature) and slowest in region A (with increasing curvature), as illustrated in Fig. 3F. Although this CNN-based model performed well in transient velocity prediction, it exhibited a limited ability in long-horizon trajectory prediction.

Multiple motion primitives induced by 2D optical patterns

A 2D composite optical-pattern terrain formed by combining multiple patterns enables a broader range of micromotor motion modes. Multiple motion primitives were developed, including propulsion within channels, in situ U-turns, passive propulsion, and motion pause and restart (Fig. 4A). Two parallel optical patterns with different widths formed a virtual channel. This channel confined and guided micromotor self-propulsion along the tangential direction of optical patterns and prevented external impurities from entering. Linear propulsion, arc propulsion, and steering around corners could be achieved in such virtual channels. The velocity of the micromotor exhibited a strong dependence on the channel height, particularly when optical patterns of large width were used (Fig. 4B). The decreased propulsion velocity indicated that the coupled local alternating electric field weakens the ICEP force within the channel. When the two light patterns had the same width, micromotors exhibited random variations in propulsion direction (Fig. 4A and movie S2) but remained confined within the channels due to the persistence of the DEP force. This effect occurred because the two optical patterns generated opposing electric fields of comparable magnitude, thereby leaving the electric field within the channel too weak to constrain the Janus interface orientation (a detailed analysis is shown in fig. S2). The removal of an optical pattern (#2) triggered micromotor propulsion along the remaining pattern (#1), with motion direction determined by the Janus interface orientation. With precise control over the timing of optical pattern removal, either in situ U-turns or straight-line motion could be selectively achieved. This strategy was effective only within specific ranges of pattern widths and channel heights; outside these ranges, in situ rotation and the subsequent U-turn could not be reliably induced, as shown in Fig. 4C. Apart from DEP and ICEP, ACEO can also play a dominant role in micromotor motion. For passive propulsion, micromotors initially guided by the narrower pattern (#1) were attracted toward the wider neighboring pattern (#2) through ACEO-induced flow. As a result, the micromotors underwent displacement perpendicular to the tangential direction of the optical pattern. The relative positions of micromotors after attraction propulsion under diverse deflection angles (φ) of two optical patterns are shown in Fig. 4D. The displacement distribution of micromotors exhibited a clear spatial bias: for $\varphi = 0$, toward the $y > x$ domain; for $\varphi = \pi/2$, near the $y = x$ line; and for $\varphi = \pi/4$, toward the $y < x$ region. In addition, micromotor motion could be stopped and restarted by projecting and then removing a U-shaped optical pattern or triangular optical pattern (fig. S3). The pattern established a 2D electric field that

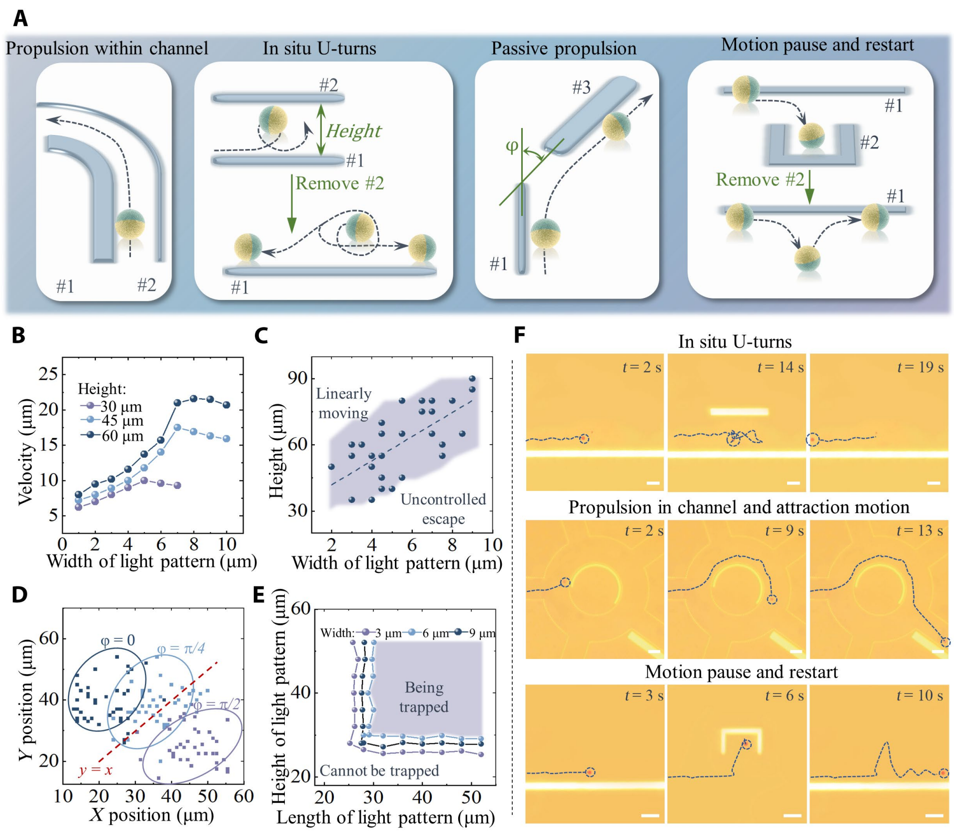

Fig. 4. Development of multiple motion primitives induced by two or more optical patterns. (**A**) Schematic illustration of proposed motion primitives, including propulsion within channels based on two parallel optical patterns, in situ U-turns based on two parallel optical patterns of consistent width, passive propulsion based on two optical patterns of diverse width and angle, and motion pause and restart based on U-shaped optical patterns. (**B**) For propulsion within channels, motion velocity as a function of both height of channels and width of the wider optical pattern. (**C**) Successful demonstrations of in situ U-turns under diverse widths and height of channels constructed by two optical patterns. (**D**) Displacement distribution of micromotors under guidance of passive propulsion versus diverse deflection angle of two optical patterns. (**E**) The critical height and length of U-shaped light patterns for motion pause and restart. (**F**) Experimental images demonstrations for in situ U-turns, propulsion within channels and passive propulsion, and motion pause and restart. Scale bar, 20 μm

attracted micromotors through ACEO flow. Stable trapping was maintained because the force-balance point between ACEO- and DEP-induced forces was within the U-shaped optical pattern (movie S3). Effective trapping within the U-shaped optical pattern required both a sufficient pattern length to guide the micromotor inward and a sufficient height to prevent escape (Fig. 4E). Four motion primitives were elucidated on the basis of three representative experiments (Fig. 4F). Micromotors achieved in situ U-turns within two optical patterns of equal width. By integrating a sequence of discrete patterns, the attraction motion primitive enabled long-term, continuous guidance of a micromotor. A micromotor from the entrance was attracted into the circular channel via the ACEO effect, self-propelled along the channel via the ICEP effect, and finally steered into the exit channel by ACEO attraction. Snapshots of the experiment that involved the "pause and restart" motion primitive are also presented in Fig. 4F. The development of these motion primitives demonstrated that the combination of ACEO, DEP, and ICEP effects broadened the flexibility and diversity of micromotor motion.

Spatial-temporal prediction based on a time-series model

Despite the high maneuverability of multiple motion primitives guided by composite optical-pattern terrains, micromotor motion under the coupled effects of DEP, ACEO, and ICEP is highly sensitive to the spatial parameters of optical patterns. As shown in figs. S4 and S5, even slight variations in the parameters of optical-pattern terrains can result in unexpected trajectories. The motion behaviors and trajectories of micromotors under guidance of composite optical-pattern terrains are challenging to predict and control as required. To address this issue, we proposed a hybrid CNN-transformer model for the motion prediction of both transient velocity and long-range motion behavior (Fig. 5A). The CNN-attention model (illustrated in Fig. 3D) was used to extract per-step spatial features of 2D optical-pattern terrains. Moreover, a transformer architecture was added to model the temporal dependencies of motion dynamics since the guidance and constraint of optical patterns also influence the time-series motion behaviors of micromotor. To improve prediction accuracy in both transient and time-series prediction tasks, we formulated a loss function. The function included four terms: a cosine loss on velocity direction, a mean squared error (MSE) loss for velocity magnitude, a unit-norm loss that penalizes deviations of the velocity-direction vector from unit length, and a 10-step accumulated position error. The ablation results are shown in Fig. 5B, which indicate that the proposed CNN-transformer consistently outperformed the benchmark models, particularly in terms of the cosine loss and accumulated position error. For a trade-off between transient and time-series prediction performance, we selected five as the optimal sequence length for the transformer model (Fig. 5C). The CNN-transformer model provides a predictive basis for accurately regulating micromotor motion across both predefined motion primitives and arbitrary optical-pattern configurations. As shown in fig. S6, the angle and amplitude of velocity predicted by the CNN-transformer model were strongly consistent with the experimental observations. For instance, the directional rotation behavior of micromotors was predicted when two optical patterns of equal width were present. In situ U-turns or straight movement were selectively triggered by modulating the removal time of one optical pattern (Fig. 5D and movie S2). Pauses and restarts of the micromotor motion primitives triggered by projecting and removing U-shaped and triangular light patterns were also predicted (Fig. 5E and movie S3). Micromotors were trapped at the central area of a T-junction formed by three wide optical patterns but could be steered out of the junction when two of the optical patterns were narrowed (Fig. 5F and movie S4). The consistency between the prediction and experimental results indicates that the CNN-transformer model effectively learned the spatiotemporal guidance and constraints induced by the 2D light patterns.

Long-term navigation with automated decision-making

The developed motion primitives, which comprised directional propulsion (A1), passive propulsion (A2), motion pause and restart (A3), and in situ U-turns (A4), enabled comprehensive omnidirectional navigation of micromotors within localized spatial domains. These motion primitives could be integrated into a discrete sequence over long-term trajectories, which were modularly reconfigurable as required (Fig. 6A). The CNN-transformer model was used to predict motion behaviors induced by predesigned optical patterns. In addition, the prediction results enabled iterative verification and optimization of both selective configuration and key geometric parameters of these motion primitives. Inappropriate parameter choices resulted in aberrant motion trajectories, as discussed previously. Among these parameters, the relative position between micromotors and optical patterns, which was independent of the 2D topography, should have been precisely controlled to ensure motion continuity. By flexibly configuring these motion primitives, distinct trajectories of micromotors were achieved within identical topographic environments. An x-shaped network that was formed by multiple optical patterns, connected four distinct workspaces, thereby enabling high-efficiency micromotor transport (Fig. 6B and movie S5). By timely switching motion primitives under appropriate spatiotemporal conditions (Fig. 6C), micromotors could be steered into arbitrarily connected channels and even returned to the original entry point. In situ U-turn motion played a dominant role when micromotors were directed toward exits No. 1 and No. 2. The experimental trajectories closely matched those predicted by the CNN-transformer model (Fig. 6D), which confirmed that the AI-based prediction and subsequent parameter optimization were effective for long-term tasks. A more intricate optical network that featured one entrance and five parallel exits was constructed (Fig. 6E). By toggling between the directional propulsion motion (A1) and the passive propulsion (A2), micromotors could be guided into any selected subchannel (Fig. 6F and movie S6). In general, A1 facilitated forward propulsion, whereas A2 enabled flexible steering. The channel-splitting network can be used for on-demand isolation and diversion of micromotors as individual micromotors can autonomously traverse through any entry channel (Fig. 6G).

In contrast to static optical patterns, dynamically sequential optical patterns demonstrated superior efficacy in multimicromotor navigation tasks, owing to reduced spatial occupancy and increased adaptability in emergent scenarios. The critical positional threshold for triggering optical pattern reconfiguration was determined through simulation tests in which a CNN-transformer predictive model was used. A deep-learning algorithm was applied for visual detection of micromotors and semantic segmentation of optical patterns (Fig. 7A), which facilitated the spatial alignment between real-time positions of micromotor and the positional threshold. Once the distances between micromotor and preset position were less than a certain threshold, optical patterns were reconfigured and the motion primitive was switched. Continuous propulsion and long-term trajectories of micromotors were achieved by automatically reconfiguring

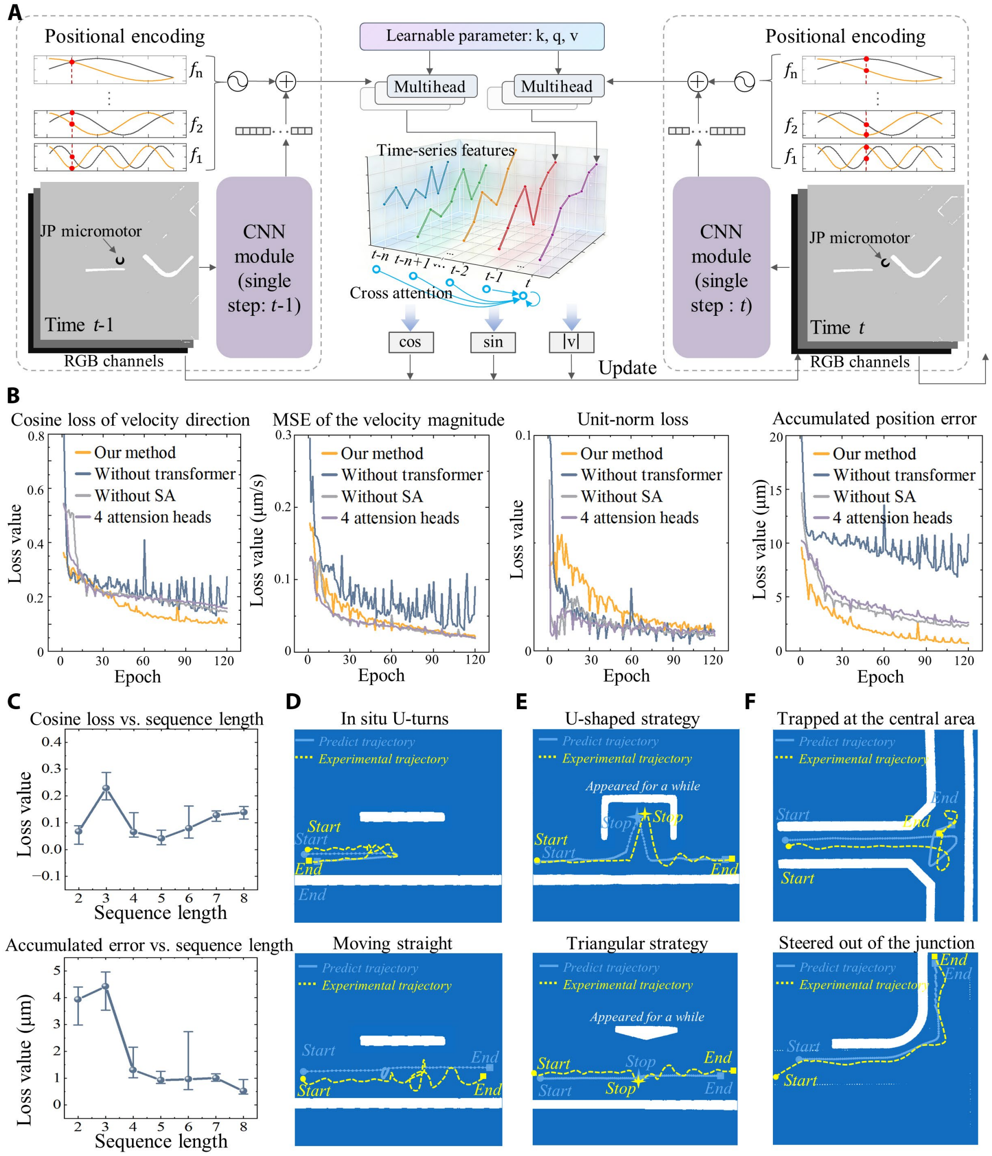

Fig. 5. Spatial-temporal prediction based on AI model. (**A**) The structure of the proposed CNN-transformer prediction model. Features extracted by CNN are subsequently processed by the time-series module after position encoding. (**B**) Performance of the proposed model compared with benchmark algorithms in terms of four components of loss function: cosine loss of velocity direction, MSE of velocity magnitude, unit-norm loss, and accumulated position error. (**C**) Cosine loss of velocity direction and accumulated position error versus sequence length in time-series module. The time-series prediction results are consistent with experimental results for navigation including (**D**) in situ U-turns, (**E**) motion pause and restart, and (**F**) motion under guidance of composite optical patterns.

optical patterns, facilitated by AI-based visual perception. To validate the efficiency of this automated reconfiguration strategy, we constructed a parking-shaped optoelectronic network with one entrance/exit and four confinement sites. This network enabled the discretized storage and release of up to four micromotors. As illustrated in Fig. 7B and movie S7, four micromotors were guided toward distinct confinement sites sequentially from the entrance/exit and then trapped by the U-shaped optical patterns. The high precision in visual recognition and semantic segmentation enabled reconfiguration of optical pattern to be triggered accurately within each critical positional threshold (Fig. 7C). Motion pause and restart primitive (A4) played a dominant role in storage of micromotors. Furthermore, micromotors were allowed to steer into and out of this network in an arbitrary order, and the use of any empty confinement sites was also order independent. For instance, micromotors were stored in the order of No. 1, No. 2, No. 3, and No. 4, and then released in the order of No. 2, No. 3, No. 4, and No. 1 (Fig. 7D).

Independent navigation of three micromotors was demonstrated within a maze-shaped optoelectronic network, which featured multiple impassable channels and only a single viable channel linking two entrances/exits (Fig. 7E and movie S8). The spatial discreteness of the optical patterns enabled localized reconfiguration of electric field while maintaining independent control over other individual micromotors. The passive motion primitive (A2) was invoked at

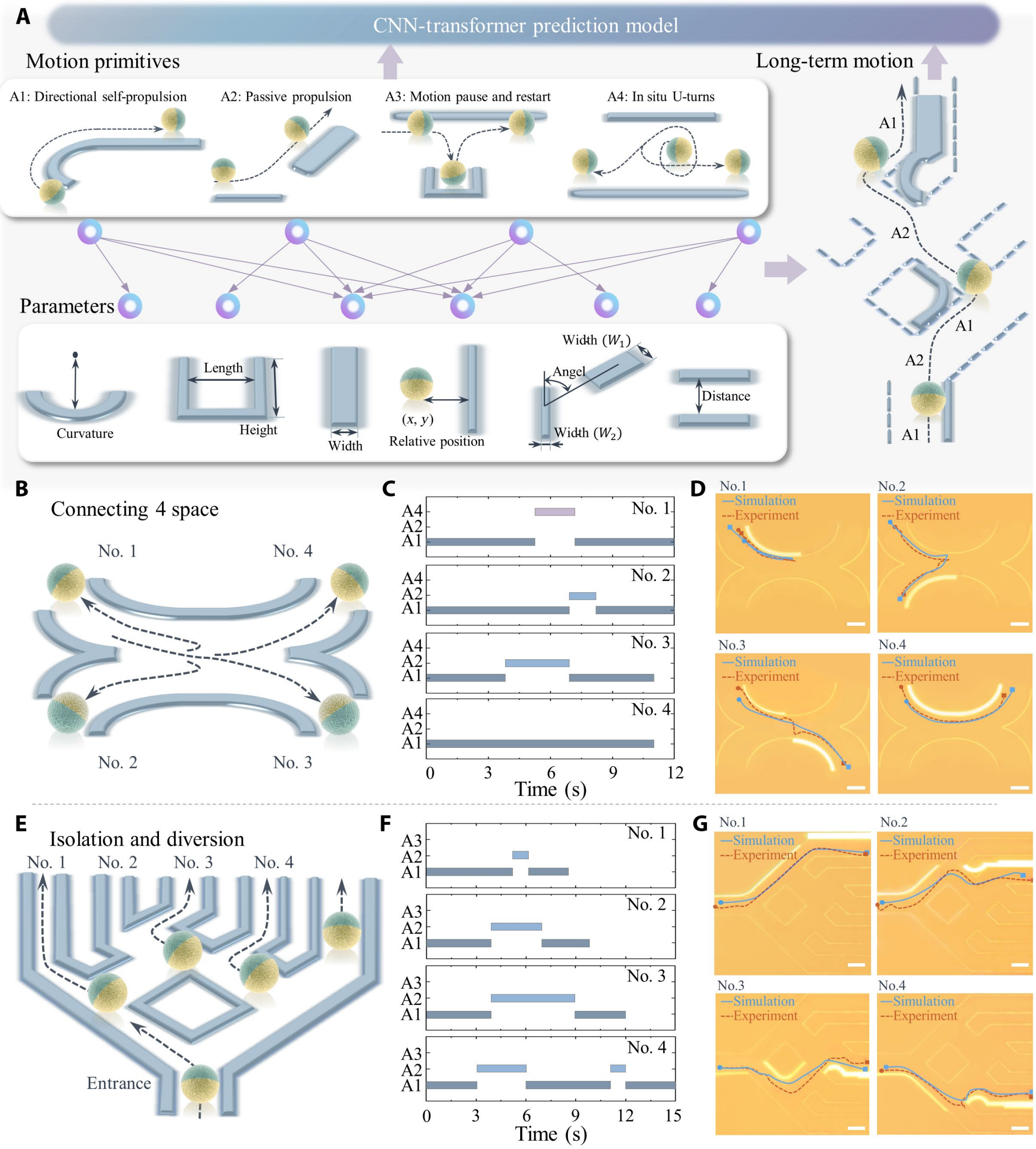

Fig. 6. Integrating motion units for long-term navigation. (**A**) Long-term navigation is guided by selectively integrating motion primitives into reconfigurable sequence. The selection of motion primitives and key parameters is iteratively optimized with the assistance of the AI-based prediction model. (**B**) In X-shaped optoelectronic network, micromotor enable steering into arbitrary channels as will. (**C**) Motion primitives of individual micromotors in X-shaped network versus time. (**D**) The consistency between predicted trajectories and experimental trajectories of micromotors steered into diverse channels. Scale bar, 30 μm. (**E**) In channel-splitting network, micromotors navigated through diverse outlets. (**F**) Motion primitives of individual micromotors in channel-splitting network versus time. (**G**) The consistency between predicted trajectories and experimental trajectories of micromotors navigation. Scale bar, 30 μm

nearly every reconfiguration step (Fig. 7F). This primitive linked the discrete motion induced by each individual optical pattern, thereby forming continuous micromotor trajectories. The representative trajectories and the real-time position traces of three micromotors are shown in Fig. 7 (G and H, respectively). At 87 s, an abrupt change in the feasible path within the arc channel occurred, which prompted immediate reconfiguration of the guiding optical pattern. The revised optical pattern successfully steered micromotor No. 3 onto a revised trajectory, thereby circumventing the obstacle. To prevent potential collisions among these actively propelling particles, the motion pause and restart primitive was used to temporarily confine selected micromotors within the U-shaped optical patterns. The selected micromotor remained confined until the feasible path was not occupied. Micromotor No. 3, which was trapped by the U-shaped optical pattern at 81 s, was subsequently released and guided to the feasible path via three sequential optical patterns within a 12-s interval (Fig. 7I).

DISCUSSION

In summary, we developed a predictable and controllable strategy for active micromotors based on optoelectronic guidance. This strategy enabled agile and precise motion of individual micromotors while

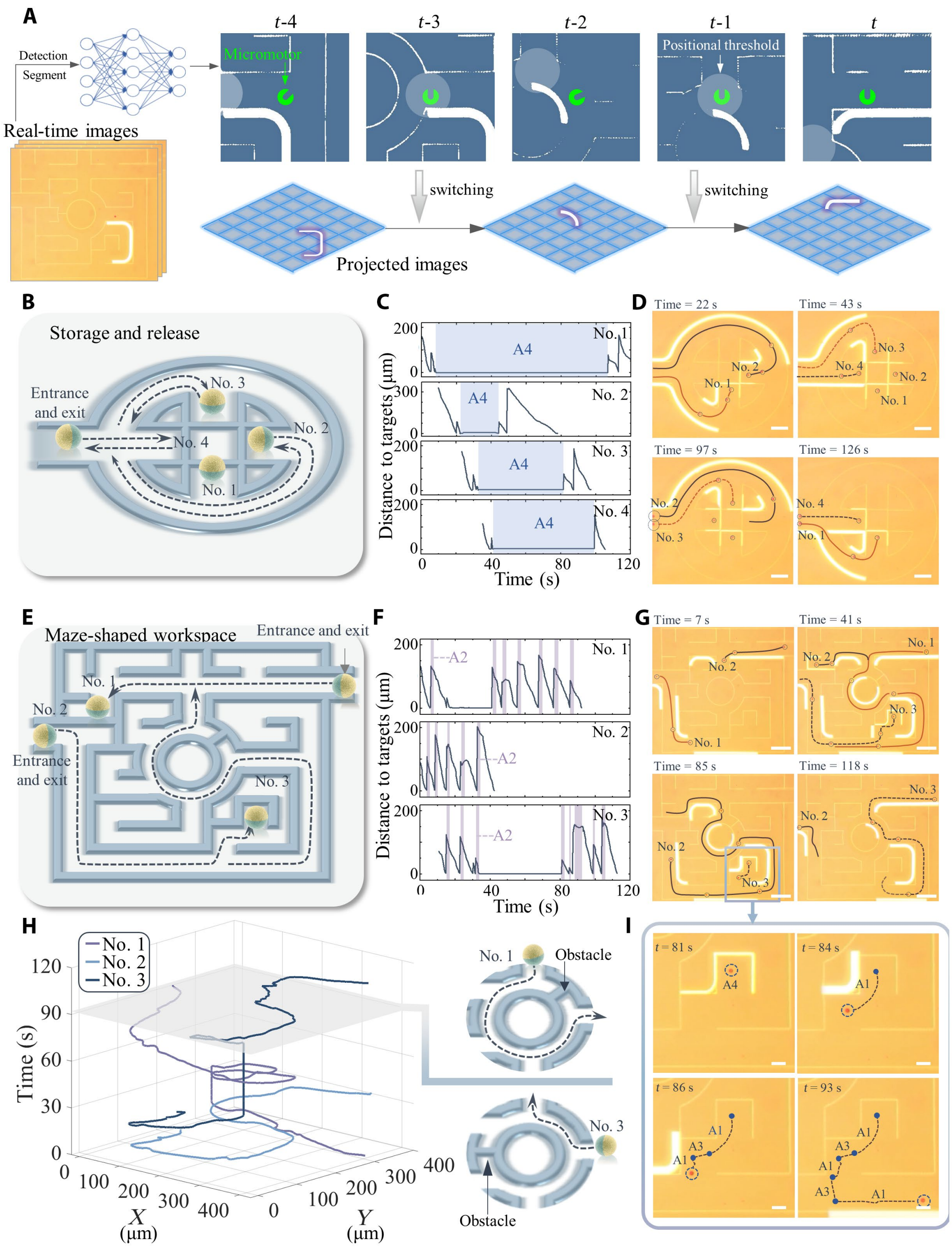

Fig. 7. AI-assisted reconfiguration of optical patterns for long-term navigation of multiple micromotors. (**A**) Automated reconfiguration of optical patterns based on real-time visual recognition and dynamic monitoring. (**B**) Storing and releasing four micromotors in a parking-shaped optoelectronic network. (**C**) Reconfiguration of optical patterns are automatedly triggered when micromotors are closest to the centers of each critical positional threshold. (**D**) Representative trajectories of four micromotors in the parking-shaped network. Scale bar, 50 μm. (**E**) Independent navigation of three micromotors in maze-shaped constraint networks. (**F**) Automated reconfiguration of optical patterns in the maze-shaped network. (**G**) Representative trajectories and (**H**) temporal evolution positions of three micromotors in the maze-shaped network. Scale bar, 50 μm. (**I**) Switching of motion primitives to navigate the No. 3 micromotor from U-shaped optical pattern to the feasible path. Scale bar, 20 μm.

supporting parallel manipulation with independent control. In contrast to conventional passive transportation and active self-propulsion, micromotor navigation was achieved in this study by leveraging spatial inhomogeneities produced by both passive (DEP and ACEO) and active (ICEP) electrokinetic mechanisms, thereby substantially increasing motion agility. Notably, omnidirectional motion, including tangential motion along optical patterns (forward and backward), normal motion toward or away from the patterns, and in-plane rotation, was achieved. Furthermore, the relative dominance of three electrokinetic effects on motion could be modulated via the spatial configuration of optical patterns. This flexible modulation allowed versatile motion primitives, including directional self-propulsion, passive propulsion, in situ U-turns, and motion pause and restart. For instance, directional steering was governed primarily by ICEP, whereas passive propulsion was actuated predominantly by ACEO.

Micromotor motion behaviors were further investigated under the coupled effect of multiple electrokinetic mechanisms and the composite guidance imposed by multiple optical patterns. Composite optical-pattern terrains enable the implementation and flexible switching of multiple micromotor motion modes, thereby improving micromotor maneuverability. However, they also introduce additional interference and complexity into micromotor dynamics under the coupled effects of three electrokinetic mechanisms. Micromotor behavior was highly sensitive to the spatial distribution of the optical-pattern terrains, as illustrated in figs. S4 and S5 and movie S9. Even small variations in the parameters could give rise to unexpected trajectories. Therefore, achieving predictable and controllable micromotor motion under coupled electrokinetic mechanisms and composite optical-pattern guidance remains highly challenging. A key advance lies in the accurate prediction of micromotor motion using a CNN-transformer model. The model learned the effects of 2D optical-pattern terrains on motion behaviors from both spatial and temporal perspectives, thereby enabling predictable and controllable micromotor motion. In the transient regime, the spatiotemporal model elucidated how optical patterns with spatially varying curvature or width induce time-varying self-propulsion velocities (movie S1). It also enabled the evaluation of the parameters of optical patterns for realizing specific motion primitives of micromotors (movies S2 to S3). In the time-series regime, the long-term trajectories of micromotors across intricate and highly composite optical-pattern terrains could be also effectively predicted via the CNN-transformer model (movies S4 to S6). With the assistance of the CNN-transformer model, micromotor motion can be accurately regulated not only across the four predefined motion modes but also under arbitrary optical-pattern configurations. To further improve the maneuverability of active micromotors during long-term navigation, sequences of discrete optical patterns were used to guide continuous micromotor motion. This strategy avoided the requirement for a single optical pattern occupying a large spatial area. The CNN-transformer model also supported the parallel manipulation and independent control of multiple micromotors since the interference arising from optical patterns associated with neighboring micromotors was incorporated into the 2D input. Simultaneous navigation of three micromotors along distinct trajectories within a maze-shaped optoelectronic network was achieved (movies S7 and S8). The advantages of the AI-assisted control strategy in maneuverability and efficiency was further validated through a collision-avoidance experiment that involved three micromotors in a confined environment with dynamic optoelectronic obstacles (fig. S7 and movie S10).

Inspired by urban traffic systems, optoelectronic networks with diverse functionalities could be constructed by integrating multiple optical patterns. These static optoelectronic networks, combined with dynamically reconfigurable optical patterns, enabled agile and controllable micromotor motions. For rapid navigation, X-shaped, crossroad-shaped, and roundabout-shaped optoelectronic networks (fig. S8) were designed to interconnect multiple working areas. Micromotors were allowed to self-propel through arbitrary entrances/exits and steer into arbitrary channels of these networks. For micromotor isolation, channel-splitting networks enabled the steering of micromotors from one entrance into multiple outlets. To store and release micromotors, a parking-shaped network provided four discrete confinement sites for effective trapping. Such orchestrated optoelectronic networks are expected to serve as templated terrains for micromotor manipulation, including parallel deployment of multiple micromotors, systematic configurations of multistep tasks, and collaboration across multiple working regions.

Beyond independent and parallel navigation of individual micromotors, we further investigated the swarm motion behaviors induced by composite 2D optical patterns. Self-aggregation (fig. S9) and self-assembly into specific micropatterns were consistently observed in experiments. For instance, micromotors from all directions were attracted to the U-shaped optical patterns via ACEO and, subsequently, self-propelled toward the interior of the U-shaped light patterns via ICEP. Programmable optical patterns were constructed as virtual molds for micromotor assembly. These patterns included linear, S-shaped, four-blade-fan–shaped, and grid-shaped architectures (fig. S10), enabling micromotors to assemble into prescribed micropatterns. Within 40 s, micromotors rapidly self-aggregated into grid-shaped assemblies (movie S1), with side lengths that exceeded 200 μm. Active micromotors in motion exhibited only transient adhesion rather than stable aggregation (fig. S11), whereas stationary micromotors had a stronger propensity for self-assembly. In future work, we will focus on investigating the local micromotor-micromotor interactions within swarms, as well as the dynamic interactions between micromotor swarms and reconfigurable optical patterns.

MATERIALS AND METHODS

Fabrication of the metallic-colloidal micromotors

PS microspheres with diameters of 5 μm (purchased from Sigma-Aldrich) in isopropanol were spin-coated onto a glass slide to form a monolayer upon solvent evaporation. Janus metallic-colloidal micromotors were fabricated by evaporating a 10-nm layer of Cr and 15-nm layer of Au on a hemisphere of PS microspheres. They were then sonicated off the substrate and dispersed in deionized water with 0.1% (v/v) Tween-20 (Sigma-Aldrich) for further use. The experimental solution was prepared by mixing 25 μM KCl (obtained by serial dilution of 1 M KCl stock solution) and 0.1% (v/v) Tween-20 (conductivity ≈5 $\mu S\ cm^{-1}$). Images of two metal-colloidal Janus micromotors that were captured by scanning electron microscope (Tescan Mira4) and energy dispersive spectroscopy (ASTM D4294) are shown in fig. S12.

Experimental setup for micromotor navigation

The optoelectronic system is illustrated in fig. S13. An optoelectronic chip with a sandwich configuration that, comprised two ITO-coated glass slides (Sigma-Aldrich) and an interposed microfluidic chamber, was used in this study. A 1-μm-thick hydrogenated amorphous silicon (a-Si:H) photosensitive layer was deposited onto the bottom ITO-coated slide via plasma-enhanced chemical vapor deposition. The height of microfluidic chamber was maintained at 70 μm by separating the two ITO slides with adhesive tape (3M). Unless otherwise specified, sinusoidal voltages with a frequency of 30 kHz and a peak-to-peak amplitude of 10 Vpp were applied to two ITO slides using a function generator (MFG-2220 HM, 25 MHz). Optical patterns that featured programmability and reconfigurability were generated by a digital micromirror device (DMD)–based projector (BenQ MH560, 3800-lumen LED source) and projected onto the a-Si:H layer through a shortpass filter (Thorlabs FGL590M, cutoff ≈ 590 nm) and a 10× objective lens (Daheng, GCO-2116).

Micromotor motion was recorded using a complementary metal-oxide semiconductor camera (STC-MCS43POE, GigE Vision) equipped with a long-pass filter (Thorlabs FELH05550, transmission region: 559 to 2150 nm). A long-pass filter was used to avoid interference from excessively bright optical patterns during observation. Dynamic light patterns were generated with a homemade GUI constructed on the basis of PyQt (version 5.15.9) and projected via the DMD-based projector.

Automated reconfiguration of the optical patterns

The automated reconfiguration of the optical patterns primarily relied on real-time visual perception of micromotors. All visual perception procedures were implemented using custom Python code (version 3.11.2) to extract key information, including position, velocity, and trajectory information, from micromotors and optical patterns. YOLOv11 (*46*) was used to localize self-propelled micromotors, and YOLOv11-seg was applied for semantic segmentation of the optical patterns. Corner points were identified from both the projected terrains and the segmentation results. By comparing their local geometric signatures, we generated a set of candidate correspondences. A robust transformation matrix was then estimated using the RANSAC algorithm to validate the correct matches and align the experimental images to the coordinates of projected terrains. For automated optical-pattern reconfiguration, trend estimation was performed by continuously evaluating the distance between each micromotor and the center of its critical positional threshold. A pattern switch was triggered when this distance shifted from a decreasing to an increasing trend. Following each reconfiguration, the system state was updated to enable continuous monitoring during subsequent navigation. The experiments were conducted on a computer configured with: i7-14700F CPU (2.10 GHz, and 64 GB RAM), NVIDIA GeForce RTX 4090 GPU. The cumulative computation speed of visual perception and spatial alignment can guarantee at least 15 frames/s. This speed satisfied the requirement for real-time optical-pattern reconfiguration.

Motion prediction

The motion prediction model adopted a composite architecture in which a CNN extracts 2D spatial features, and the transformer module serves as the temporal feature extractor. This design allowed for synergistic learning of high-level features from both spatial information and the micromotors' historical motion. ResNet blocks, which have been extensively validated across a wide range of computer vision tasks, are adopted as the backbone of the CNN model. Spatial attention was implemented by computing a weight matrix and applying it to the intermediate CNN feature maps. This mechanism enabled the model to effectively extract spatial features of optical pattern terrains (fig. S14). The transformer model used eight attention heads, which represented a trade-off between the predictive performance and computational cost. The proposed CNN model pretrained for the transient prediction task, and its weights were subsequently adopted as initial parameters by the composite model to accelerate convergence and increase predictive accuracy. Model training follows a hybrid strategy that integrates synthetic and experimental data. Synthetic data generated primarily from statistical models, thereby enabling improved generalization across diverse optical-pattern parameters. Extensive comparative experiments (table S2), and evaluations against multiple architectures such as a YOLOv11-transformer model, were conducted to validate the superiority of the proposed model. The positional errors between the long-term trajectories predicted by the CNN-transformer model and the experimental trajectories are shown in fig. S15.

Numerical simulation

FEA was performed using the commercial software COMSOL Multiphysics (version 5.6, COMSOL Inc.) to investigate the dynamics of the micromotors under optical pattern–induced guidance and confinement. A 3D FEA model that comprised a 15-μm-thick ITO layer, a 1-μm-thick a-Si:H layer, and 70-μm-thick surrounding liquid medium was constructed and meshed in a superfine element manner. The thicknesses of the Cr (10 nm) and Au (15 nm) coating in the FEA analysis remained consistent with those in the experiments. The conductivity of the illuminated regions was set to be higher than that of the dark regions. Numerical simulations of the DEP and ACEO effects were carried out using both the electric currents and microfluidics modules.

Theoretical basis of DEP and ACEO

DEP refers to the electrokinetic response of a nonuniformly polarized particle exposed to a spatially electric field, which was generated by projected optical patterns rather than conventional metal electrodes in this study. The direction of the DEP force is strongly influenced by the electric field frequency (*47*, *48*), the electrical properties of the liquid medium, and the geometry and size of microparticles. In this study, the thicknesses of the Cr and Au coatings were small, which resulted in n-DEP and the release of micromotors from restricted confinement within the optical pattern by positive DEP (p-DEP). According to the Lorentz law, the momentum changes of a micromotor within electromagnetic fields can be expressed as follows

$$\frac{d}{d_t}\left(\mathbf{P_M}+\mathbf{P_F}\right) = \oint_S (T \cdot n)dS \tag{1}$$

where $\mathbf{P_M}$ represents the momentum of the micromotor, $\mathbf{P_F}$ is the total momentum of electromagnetic field, S is the enclosed surface of the micromotor, n is the unit tensor, and T is the Maxwell stress tensor.

ACEO refers to the fluid motion induced at the interface between a liquid and a charged surface under a tangential alternating electric field (*49*), driven by electrostatic interactions with ions in the electrical double layer. To leverage the effect of ACEO in propelling micromotors toward optical patterns, the frequency of the applied alternating electric field was set to 30 kHz. The velocity of the electrical double layer actuated by ACEO can be expressed as follows

$$\mathbf{v_{ACEO}} = \frac{1}{8}\frac{\varepsilon V_0^2 \Omega^2}{\mu x\left(1+\Omega^2\right)^2}\frac{C_S}{C_S + C_D} \tag{2}$$

where Ω is the nondimensional frequency related to the AC electric frequency, medium permittivity and conductivity; C_S is the capacitance of the Stern layer; $C_D$represents the capacitance of the diffuse layer, and V_0 is the potential value of the applied AC electric field.

Supplementary Materials

The PDF file includes:

Figs. S1 to S15

Tables S1 and S2

Legends for movies S1 to S10

Other Supplementary Material for this manuscript includes the following:
Movies S1 to S10

REFERENCES

1. Z. Ghalanbor, S.-A. Marashi, B. Ranjbar, Nanotechnology helps medicine: Nanoscale swimmers and their future applications. *Med. Hypotheses* **65**, 198–199 (2005).
2. R. Lipowsky, Y. Chai, S. Klumpp, S. Liepelt, M. J. I. Müller, Molecular motor traffic: From biological nanomachines to macroscopic transport. *Phys. A Stat. Mech. Appl.* **372**, 34–51 (2006).
3. P. Degen, Self-propelling capsules as artificial microswimmers. *Curr. Opin. Colloid Interface Sci.* **19**, 611–619 (2014).
4. L. Alvarez, M. A. Fernandez-Rodriguez, A. Alegria, S. Arrese-Igor, K. Zhao, M. Kröger, L. Isa, Reconfigurable artificial microswimmers with internal feedback. *Nat. Commun.* **12**, 4762 (2021).
5. J. Katuri, X. Ma, M. M. Stanton, S. Sánchez, Designing micro-and nanoswimmers for specific applications. *Acc. Chem. Res.* **50**, 2–11 (2017).
6. W. Liu, X. Chen, X. Lu, J. Wang, Y. Zhang, Z. Gu, From passive inorganic oxides to active matters of micro/nanomotors. *Adv. Funct. Mater.* **30**, 2003195 (2020).
7. M. Fränzl, S. Muinos-Landin, V. Holubec, F. Cichos, Fully steerable symmetric thermoplasmonic microswimmers. *ACS Nano* **15**, 3434–3440 (2021).
8. S. A. Mallory, C. Valeriani, A. Cacciuto, An active approach to colloidal self-assembly. *Annu. Rev. Phys. Chem.* **69**, 59–79 (2018).
9. Q. Yang, L. Xu, W. Zhong, Q. Yan, Y. Gao, W. Hong, Y. She, G. Yang, Recent advances in motion control of micro/nanomotors. *Adv. Intell. Syst.* **2**, 2000049 (2020).
10. Y. Hu, W. Liu, Y. Sun, Self-propelled micro−/nanomotors as "on-the-move" platforms: Cleaners, sensors, and reactors. *Adv. Funct. Mater.* **32**, 2109181 (2022).
11. A. F. Demirörs, F. Eichenseher, M. J. Loessner, A. R. Studart, Colloidal shuttles for programmable cargo transport. *Nat. Commun.* **8**, 1872 (2017).
12. D. Xu, Y. Wang, C. Liang, Y. You, S. Sanchez, X. Ma, Self-propelled micro/nanomotors for on-demand biomedical cargo transportation. *Small* **16**, 1902464 (2020).
13. S. Erez, E. Karshalev, Y. Wu, J. Wang, G. Yossifon, Electrical propulsion and cargo transport of microbowl shaped Janus particles. *Small* **18**, 2101809 (2022).
14. X. Lin, H. Zhu, Z. Zhao, C. You, Y. Kong, Y. Zhao, J. Liu, H. Chen, X. Shi, D. Makarov, Y. Mei, Hydrogel-based Janus micromotors capped with functional nanoparticles for environmental applications. *Adv. Mater. Technol.* **5**, 2000279 (2020).
15. S. Park, G. Yossifon, Micromotor-based biosensing using directed transport of functionalized beads. *ACS Sens.* **5**, 936–942 (2020).
16. Z. Rahiminezhad, A. M. Tamaddon, S. Borandeh, S. S. Abolmaali, Janus nanoparticles: New generation of multifunctional nanocarriers in drug delivery, bioimaging and theranostics. *Appl. Mater. Today* **18**, 100513 (2020).
17. V. Milosavljevic, L. Kosaristanova, K. Dolezelikova, V. Adam, M. Pumera, Microrobots with antimicrobial peptide nanoarchitectonics for the eradication of antibiotic-resistant biofilms. *Adv. Funct. Mater.* **32**, 2112935 (2022).
18. W. Wang, C. Zhou, A journey of nanomotors for targeted cancer therapy: Principles, challenges, and a critical review of the state-of-the-art. *Adv. Healthc. Mater.* **10**, 2001236 (2021).
19. Y. Ying, M. Pumera, Micro/nanomotors for water purification. *Chem. Eur. J.* **25**, 106–121 (2019).
20. M. S. Draz, K. M. Kochehbyoki, A. Vasan, D. Battalapalli, A. Sreeram, M. K. Kanakasabapathy, S. Kallakuri, A. Tsibris, D. R. Kuritzkes, H. Shafiee, DNA engineered micromotors powered by metal nanoparticles for motion based cellphone diagnostics. *Nat. Commun.* **9**, 4282 (2018).
21. A. Poddar, A. Bandopadhyay, S. Chakraborty, Steering a thermally activated micromotor with a nearby isothermal wall. *J. Fluid Mech.* **915**, A22 (2021).
22. A. Al Harraq, M. Bello, B. Bharti, A guide to design the trajectory of active particles: From fundamentals to applications. *Curr. Opin. Colloid Interface Sci.* **61**, 101612 (2022).
23. F. Mou, Q. Xie, J. Liu, S. Che, L. Bahmane, M. You, J. Guan, ZnO-based micromotors fueled by CO_2: The first example of self-reorientation-induced biomimetic chemotaxis. *Natl. Sci. Rev.* **8**, nwab066 (2021).
24. C. Lozano, B. Ten Hagen, H. Löwen, C. Bechinger, Phototaxis of synthetic microswimmers in optical landscapes. *Nat. Commun.* **7**, 12828 (2016).
25. G. Tkachenko, V. G. Truong, C. L. Esporlas, I. Sanskriti, S. Nic Chormaic, Evanescent field trapping and propulsion of Janus particles along optical nanofibers. *Nat. Commun.* **14**, 1691 (2023).
26. C. Chen, F. Mou, L. Xu, S. Wang, J. Guan, Z. Feng, Q. Wang, L. Kong, W. Li, J. Wang, Q. Zhang, Light-steered isotropic semiconductor micromotors. *Adv. Mater.* **29**, 1603374 (2017).
27. J. Palacci, S. Sacanna, A. Abramian, J. Barral, K. Hanson, A. Y. Grosberg, D. J. Pine, P. M. Chaikin, Artificial rheotaxis. *Sci. Adv.* **1**, e1400214 (2015).
28. C. Gao, Y. Feng, D. A. Wilson, Y. Tu, F. Peng, Micro-nano motors with taxis behavior: Principles, designs, and biomedical applications. *Small* **18**, 2106263 (2022).
29. X. Xia, Y. Li, X. Xiao, Z. Zhang, C. Mao, T. Li, M. Wan, Chemotactic micro/nanomotors for biomedical applications. *Small* **20**, 2306191 (2024).
30. D. Zhou, R. Zhuang, X. Chang, L. Li, Enhanced light-harvesting efficiency and adaptation: A review on visible-light-driven micro/nanomotors. *Research* **2020**, 6821595 (2020).
31. M. You, C. Chen, L. Xu, F. Mou, J. Guan, Intelligent micro/nanomotors with taxis. *Acc. Chem. Res.* **51**, 3006–3014 (2018).
32. S. Das, A. Garg, A. I. Campbell, J. Howse, A. Sen, D. Velegol, R. Golestanian, S. J. Ebbens, Boundaries can steer active Janus spheres. *Nat. Commun.* **6**, 8999 (2015).
33. J. Simmchen, J. Katuri, W. E. Uspal, M. N. Popescu, M. Tasinkevych, S. Sánchez, Topographical pathways guide chemical microswimmers. *Nat. Commun.* **7**, 10598 (2016).
34. J. Katuri, D. Caballero, R. Voituriez, J. Samitier, S. Sánchez, Directed flow of micromotors through alignment interactions with micropatterned ratchets. *ACS Nano* **12**, 7282–7291 (2018).
35. L. Zhang, Z. Xiao, X. Chen, J. Chen, W. Wang, Confined 1D propulsion of metallodielectric Janus micromotors on microelectrodes under alternating current electric fields. *ACS Nano* **13**, 8842–8853 (2019).
36. C. Liu, C. Zhou, W. Wang, H. P. Zhang, Bimetallic microswimmers speed up in confining channels. *Phys. Rev. Lett.* **117**, 198001 (2016).
37. F. Yang, F. Mou, Y. Jiang, M. Luo, L. Xu, H. Ma, J. Guan, Flexible guidance of microengines by dynamic topographical pathways in ferrofluids. *ACS Nano* **12**, 6668–6676 (2018).
38. T. Xu, C. Huang, Z. Lai, X. Wu, Independent control strategy of multiple magnetic flexible millirobots for position control and path following. *IEEE Trans. Robot.* **38**, 2875–2887 (2022).
39. Q. Gao, Z. Yang, R. Zhu, J. Wang, P. Xu, J. Liu, X. Chen, Z. Yan, Y. Peng, Y. Wang, H. Zheng, F. Cai, W. Wang, Ultrasonic steering wheels: Turning micromotors by localized acoustic microstreaming. *ACS Nano* **17**, 4729–4739 (2023).
40. T. Luo, M. Wu, Biologically inspired micro-robotic swimmers remotely controlled by ultrasound waves. *Lab Chip* **21**, 4095–4103 (2021).
41. P. Y. Chiou, A. T. Ohta, M. C. Wu, Massively parallel manipulation of single cells and microparticles using optical images. *Nature* **436**, 370–372 (2005).
42. S. S. Das, G. Yossifon, Optoelectronic trajectory reconfiguration and directed self-assembly of self-propelling electrically powered active particles. *Adv. Sci.* **10**, 2206183 (2023).
43. X. Chen, X. Chen, M. Elsayed, H. Edwards, J. Liu, Y. Peng, H. P. Zhang, S. Zhang, W. Wang, A. R. Wheeler, Steering micromotors via reprogrammable optoelectronic paths. *ACS Nano* **17**, 5894–5904 (2023).
44. F. He, T. Liu, D. Tao, Why resnet works? residuals generalize. *IEEE Trans. Neural Networks Learn. Syst.* **31**, 5349–5362 (2020).
45. M. Zhu, L. Jiao, F. Liu, S. Yang, J. Wang, Residual spectral–spatial attention network for hyperspectral image classification. *IEEE Trans. Geosci. Remote Sens.* **59**, 449–462 (2020).
46. R. Khanam, M. Hussain, Yolov11: An overview of the key architectural enhancements. arXiv:2410.17725 [cs.CV] (2024).
47. X. Hong, B. Xu, G. Li, F. Nan, X. Wang, Q. Liang, W. Dong, W. Dong, H. Sun, Y. Zhang, Optoelectronically navigated nano-kirigami microrotors. *Sci. Adv.* **10**, eadn7582 (2024).
48. Y. Zhang, J. Zhao, H. Yu, P. Li, W. Liang, Z. Liu, G.-B. Lee, L. Liu, W. J. Li, Z. Wang, Detection and isolation of free cancer cells from ascites and peritoneal lavages using optically induced electrokinetics (OEK). *Sci. Adv.* **6**, eaba9628 (2020).
49. D. Das, Z. Yan, N. V. Menon, Y. Kang, V. Chan, C. Yang, Continuous detection of trace level concentration of oil droplets in water using microfluidic AC electroosmosis (ACEO). *RSC Adv.* **5**, 70197–70203 (2015).

Acknowledgments: We thank C. Li and Z. Li for assistance in active micromotor fabrication. **Funding:** This work was supported by the National Natural Science Foundation of China under grant number 62573060 (H.W.), National Natural Science Foundation of China under grant number 62403056 (Y.H.), Beijing Natural Science Foundation under grant L242023 (H.W.), and Fundamental Research Funds for the Central Universities under grant 2025CX01003 (H.W.). **Author contributions:** Conceptualization: J.L. and Z.Z. Methodology: J.L. and Z.Z. Investigation: J.L., Z.Z., Q.S., and Q.H. Funding acquisition: H.W. and M.S. Supervision: H.W., M.S., Q.S., and Q.H. Writing—original draft: J.L., Z.Z., J.H., H.W., and M.S. Writing—review and editing: Y.H., H.W., and M.S. **Competing interests:** The authors declare that they have no competing interests. **Data, code, and materials availability:** All data needed to evaluate and reproduce the results in the paper are present in the paper and/or the Supplementary Materials. The source code is fully available at https://github.com/husandaimei/ResNet-Transformer-for-motion-prediction.git and https://doi.org/10.5281/zenodo.19245390. This study did not generate new materials.

Submitted 28 December 2025
Accepted 8 July 2026
Published 14 August 2026
10.1126/sciadv.aef0741

Science Advances

AI-assisted optoelectrokinetic control of active self-propelling micromotors for independent navigation with multimode motions

Jiaxin Liu, Zhiqiang Zheng, Yaozhen Hou, Qing Shi, Qiang Huang, Jie Han, Metin Sitti, and Huaping Wang

Sci. Adv. **12** (33), eaef0741. DOI: 10.1126/sciadv.aef0741

Science Advances (ISSN 2375-2548) is published by the American Association for the Advancement of Science. 1200 New York Avenue NW, Washington, DC 20005. The title *Science Advances* is a registered trademark of AAAS.